

Common Fixed Points & Sequences

Chapter 5e: Pathak's Fixed Point Theory

Nonlinear Analysis & Fixed Point Theory

2024

Overview

- 1 5.5 Common Fixed Point Theorems
 - 5.5.2 Families of Mappings
- 2 5.6 Sequences of Contractions
- 3 5.7 Fixed Point Theorems in Ordered Banach Spaces
- 4 Summary and Connections

Section 5.5: Common Fixed Point Theorems

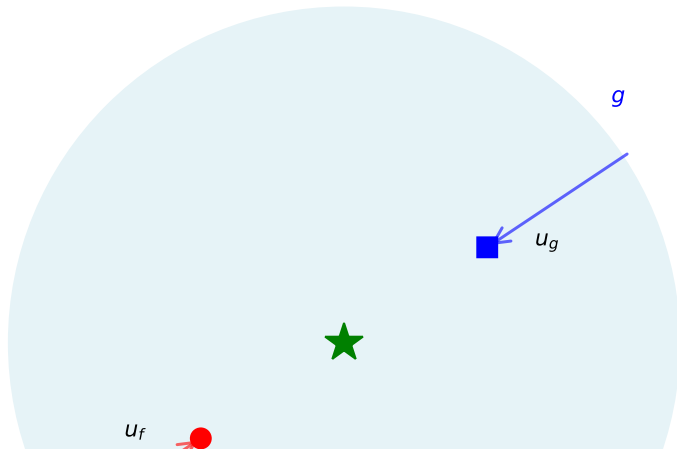
- **Definition:** A point $u \in X$ is a **common fixed point** of mappings f and g if:

$$f(u) = u \quad \text{and} \quad g(u) = u$$

- **Key Question:** When do two (or more) mappings have a common fixed point?
- **History:**
 - Markov (1936): First result on common fixed points
 - Kakutani (1938): Direct proof using Tychonoff's theorem
 - Subsequent extensions for families of mappings
- **Applications:**
 - Systems of equations
 - Simultaneous fixed points of operators
 - Optimization problems

Common Fixed Points Visualization

- Fixed point of f
 - Fixed point of g
 - ★ Common fixed point $u = f(u) = g(u)$
- Space X



Theorem 5.118: Two Mappings with Weak Compatibility

Theorem 5.118 (Weakly Compatible Mappings): Let (X, d) be a b-metric space and let $f, g : X \rightarrow X$ be mappings satisfying:

$$d(f(x_1, x_2, \dots, x_k), g(x_2, x_3, \dots, x_{k+1})) \leq \lambda \phi(d(gx_1, gx_2), \dots)$$

where $\lambda \in (0, \frac{1}{s})$, $s \geq 1$, and $g(X)$ is complete.

If f and g are **weakly compatible**, then:

- 1 $C(f, g) \neq \emptyset$ (common fixed point exists)
- 2 Sequences defined by iteration converge to the common fixed point
- 3 The common fixed point is unique

Definition (Weak Compatibility): f and g are weakly compatible if they commute at their common fixed points.

Example 5.47: Concrete Common Fixed Points

Example 5.47: Let $X = \mathbb{R}$, $d(x, y) = |x - y|^3$, $s = 4$. Define:

$$f(x, y) = \frac{x^2 + y^2}{13} + \frac{18}{13}, \quad g(x) = x^2 - 2$$

Verification:

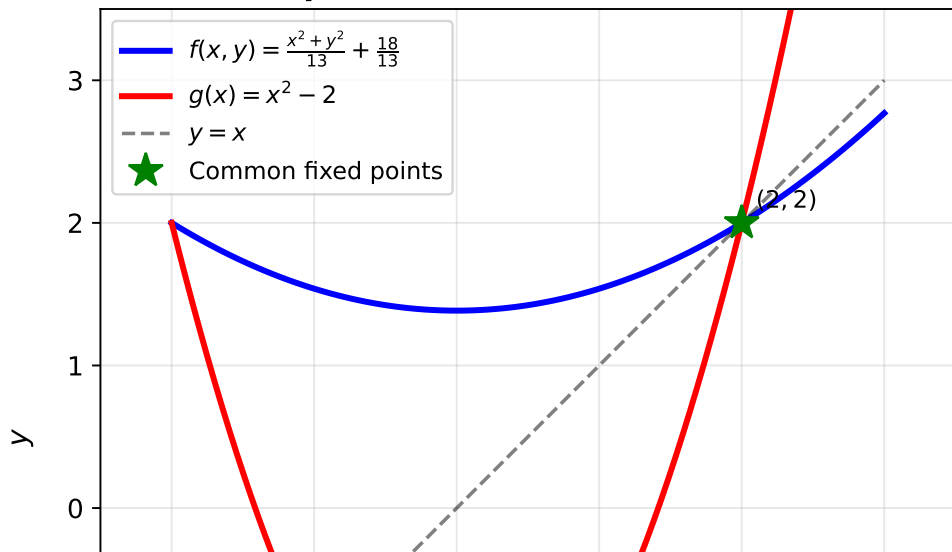
- Check condition (5.115): $d(f(x, y), f(y, z)) \leq \lambda \phi(d(gx, gy), \dots)$
- With $\lambda = \frac{8}{13^3} \in (0, \frac{1}{4})$
- Functions f and g commute at fixed points

Result: Common fixed point is $u = 2$, where:

$$f(-1, -1) = 1 = (-1)^2 - 2, \quad f(1, 1) = 2 = 2^2 - 2$$

Example 5.47 Visualization

Example 5.47: Common Fixed Points



5.5.2: Common Fixed Points of Family of Mappings

Definition 5.67: Let K be a convex subset of a normed linear space X and let F be a self-mapping of K . F is **affine mapping** if:

$$F(\alpha x + (1 - \alpha)y) = \alpha Fx + (1 - \alpha)Fy \quad \forall x, y \in K, 0 < \alpha < 1$$

Definition 5.68: A family $\mathcal{F}$ of linear mappings is **equicontinuous on subset K** if:

For every neighborhood V of the origin and $\delta > 0$, there exists neighborhood U such that if $x - y \in U$, then $F(x) - F(y) \in V$ for all $F \in \mathcal{F}$.

Key Theorem (Markov-Kakutani 1936-1938):

Compact convex subset + commuting affine maps $\Rightarrow$ non-empty common fixed point set

Theorem 5.120: Markov-Kakutani Theorem

Theorem 5.120 (Markov-Kakutani): Let K be a compact convex subset of a normed linear space X . Let $\mathcal{F}$ be a commuting family of continuous affine mappings which map K into itself.

Then there exists a point $u \in K$ such that $Fu = u$ for each $F \in \mathcal{F}$.

Proof Strategy:

- 1 Define $F_n(K) = n^{-1}(I + F + \cdots + F^{n-1})(K)$
- 2 Show $F_n(K)$ are convex and compact
- 3 Finite subfamily has non-void intersection (compactness)
- 4 Common intersection point is fixed by all $F \in \mathcal{F}$

Theorem 5.121: Kakutani's Equicontinuous Extension

Theorem 5.121 (Kakutani [302]): Let K be a compact convex subset of a normed linear space X , and let $\mathcal{F}$ be a group of affine mappings which is equicontinuous on K and such that $F(K) \subset K$ for all $F \in \mathcal{F}$.

Then there exists a point $u \in K$ such that $Fu = u$ for all $F \in \mathcal{F}$.

Key Differences from Theorem 5.120:

- Requires equicontinuity instead of commutativity
- Group structure (all mappings invertible)
- More general setting

Theorems 5.122-5.129: Nonexpansive Mappings

Theorem 5.122 (Ryll-Nadzewski [532]): Extension for semigroups of nonexpanding mappings

Theorem 5.123 (de Marr [165]): Family of commuting nonexpansive mappings on compact convex Banach space subset has common fixed point.

Theorem 5.124 (Belluce-Kirk [46]): Closed, bounded, convex subset + commuting nonexpansive family $\Rightarrow$ common fixed point

Theorem 5.125 (Weakly Compact Generalization): For weakly compact convex subsets of strictly convex Banach spaces

Theorem 5.126 (Browder [96]): Uniformly convex Banach spaces

Theorem 5.127: Multivalued mappings with generalized Kannan-Reich condition

Theorem 5.128: Sequences of self-mappings with generalized Kannan-Reich type condition

Theorem 5.129: Continuous function ϕ in multiple variables

Iteration Schemes for Families

Theorem 5.130-5.131: Iteration schemes converging to common fixed points

For finite family $\{F_i\}_{i=1}^k$ of nonexpansive mappings on compact convex Banach space, define:

$$U_0 = I \quad (\text{identity mapping})$$

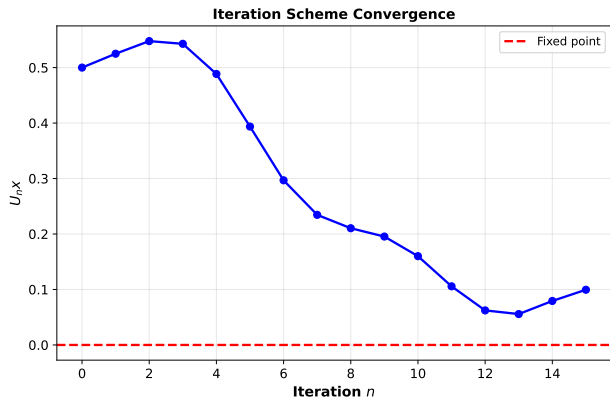
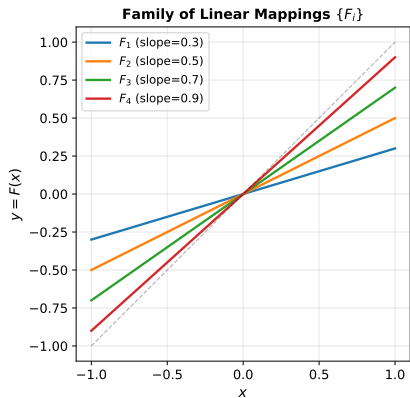
$$U_r = (1 - \alpha)I + \alpha F_r U_{r-1}, \quad r = 1, 2, \dots, k$$

Results:

- For $0 < \alpha < 1$, sequence $\{U_n^k x\}$ converges strongly to common fixed point
- Weak convergence in Hilbert/reflexive spaces
- Iterations reduce to solving systems of equations: $x - T_i x = f_i$

Remark 5.49: Application to systems of equations in Banach spaces

Family of Mappings Visualization



Section 5.6: Sequences of Contractions

Main Question: If we have a sequence of contraction mappings $\{T_n\}$ converging to a mapping T , does the sequence of fixed points converge to the fixed point of T ?

Two Types of Convergence:

- ① **Pointwise Convergence:** $\lim_{n \rightarrow \infty} T_n x = T x$ for each x
- ② **Uniform Convergence:** $\lim_{n \rightarrow \infty} d(T_n x, T x) = 0$ uniformly

Key Results:

- Bonsal [64]: Fixed point continuity for contractions
- Nadler Jr. [417, 416]: Behavior of fixed points for set-valued maps
- Conditions for convergence of fixed point sequences

Theorem 5.133: Contraction Sequences (Bonsal)

Theorem 5.133 (Bonsal [64]): Let (X, d) be a complete metric space, and let T and T_n ($n = 1, 2, \dots$) be contraction mappings of X into itself with the same Lipschitz constant $k < 1$, with fixed points u and u_n , respectively.

Suppose that $\lim_{n \rightarrow \infty} T_n x = Tx$ for every $x \in X$.

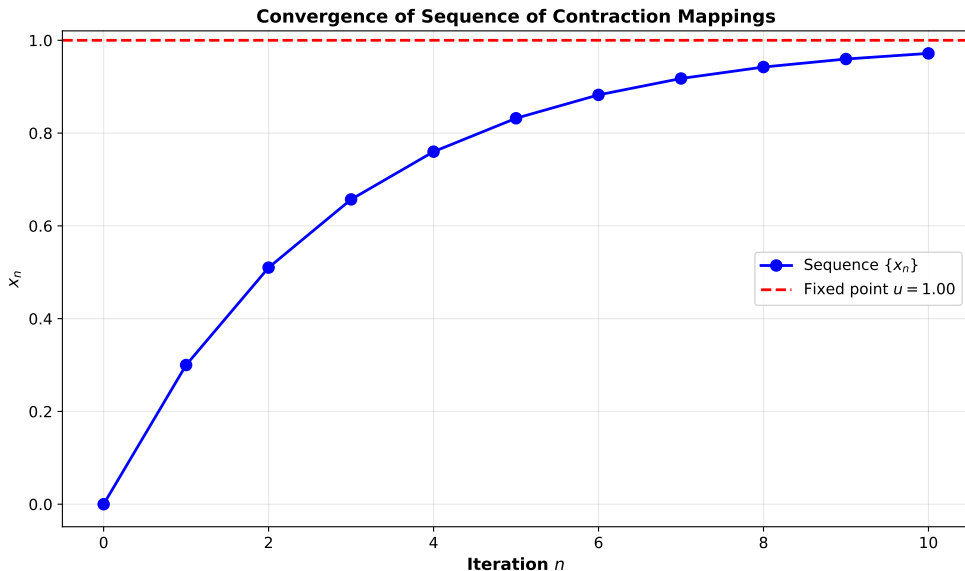
Then: $\lim_{n \rightarrow \infty} u_n = u$

Proof Sketch:

$$d(u_r, T_r^n x_0) \leq \frac{k^n}{1-k} d(Tx_0, x_0)$$

Combined with pointwise convergence gives convergence of fixed points.

Contraction Convergence Visualization



Theorem 5.134-5.135: Uniform Convergence

Theorem 5.134: Let (X, d) be a metric space. Let $T_i : X \rightarrow X$ be a mapping with at least one fixed point x_i for each $i = 1, 2, \dots$, and let $T_0 : X \rightarrow X$ be a contraction mapping with fixed point x_0 . If the sequence $\{T_i\}$ converges uniformly to T_0 , then the sequence $\{x_i\}$ converges to x_0 .

Theorem 5.135: Let (X, d) be a locally compact metric space. Let $T_i : X \rightarrow X$ be a contraction mapping with fixed point x_i for each $i = 1, 2, \dots$, and let $T_0 : X \rightarrow X$ be a contraction mapping with fixed point x_0 . If the sequence $\{T_i\}$ is equicontinuous and converges pointwise to T_0 , then the sequence $\{x_i\}$ converges to x_0 .

Theorem 5.136: Finite Dimensionality Characterization

Theorem 5.136 (Nadler Jr. [417]): A separable or reflexive Banach space is finite dimensional if and only if, whenever for a pointwise convergent sequence of contraction mappings, the sequence of fixed point converges to the fixed point of the pointwise limit mapping.

Corollary: This provides a topological characterization of finite dimensional spaces through fixed point behavior.

Interpretation:

- In finite dimensions: convergence of mappings $\Rightarrow$ convergence of fixed points
- In infinite dimensions: may fail without additional conditions
- Shows fundamental difference between finite and infinite dimensional spaces

Theorem 5.137-5.138: Multivalued Contractions

Theorem 5.137 (Nadler Jr. [416]): Let (X, d) be a complete metric space. Let $T_i : X \rightarrow \mathcal{K}(X)$ be multivalued contraction mappings with fixed point x_i for each $i = 1, 2, \dots$, and let $T_0 : X \rightarrow \mathcal{K}(X)$ be a multivalued contraction mapping.

If (i) each T_i has Lipschitz constant $\eta < 1$, (ii) $\{T_i\}$ converges uniformly to T_0 , or (iii) (X, d) is locally compact and $\{T_i\}$ converges pointwise to T_0 , then there is a subsequence $\{x_j\}$ that converges to a fixed point of T_0 .

Theorem 5.138 (Markin [385]): For setvalued contractions with Hausdorff metric, fixed point sets converge in Hausdorff metric.

Theorems 5.139-5.142: Weak Convergence

Theorem 5.139 (Markin [386]): Weak convergence of fixed points for setvalued nonexpansive mappings in strictly convex Banach spaces with weakly continuous duality map

Definition 5.69: A mapping $T : X \rightarrow X$ is of **generalized Kannan-Reich type** if:

$$d(Tx, Ty) \leq a_1 d(Tx, x) + a_2 d(Ty, y) + a_3 d(Ty, x) + a_4 d(Tx, y) + a_5 d(x, y)$$

where $\sum_{i=1}^5 a_i < 1$ and $a_1 = a_2, a_3 = a_4$

Theorem 5.140: Uniform convergence of Kannan-Reich type mappings preserves fixed points

Theorem 5.141: Pointwise convergence with Kannan-Reich property

Theorem 5.142: Multivalued mappings with similar generalized conditions

Theorem 5.143: Husain-Sehgal Type Mappings

Theorem 5.143: Let T and a sequence $\{T_n\}$ be self-mappings of a complete metric space X such that $T_n \rightarrow T$ uniformly.

Suppose for each $n \geq 1$, T_n has a fixed point x_n and T is a mapping of **Husain-Sehgal type**.

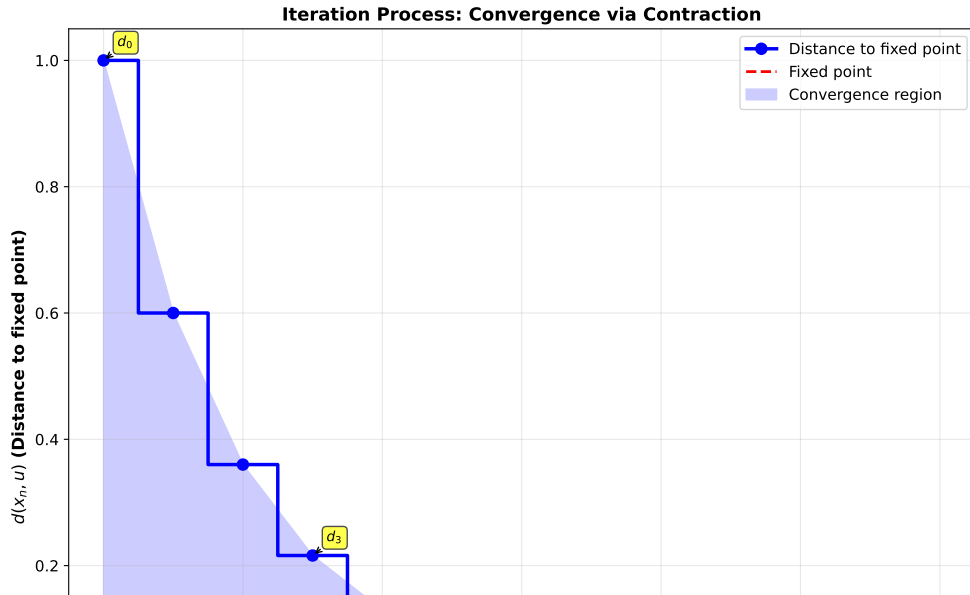
If u is the fixed point of T and $\sup d(x_n, u) < \infty$, then $x_n \rightarrow u$.

Husain-Sehgal Type Condition: Involves $(I - T)$ being J-monotone

Applications:

- Differential equations with perturbations
- Stability of fixed points under approximation
- Numerical methods with rounding errors

Iteration Process Visualization



Section 5.7: Ordered Banach Spaces

Definition 5.70: Let X be a real Banach space. A set $K \subset X$ is a **cone** if:

- 1 K is closed
- 2 If $u, v \in K$, then $\alpha u + \beta v \in K$ for all $\alpha, \beta \geq 0$
- 3 $x, -x \in K \Rightarrow x = 0$

Definition 5.71: A linear space X is **partially ordered** by relation $\leq$ if:

- 1 $x \leq y \Rightarrow tx \leq ty$ for $t \geq 0$
- 2 $x \leq y$ and $y \leq x$ implies $x = y$
- 3 $x_1 \leq y_1$ and $x_2 \leq y_2$ implies $x_1 + x_2 \leq y_1 + y_2$
- 4 $x \leq y$ and $y \leq z$ implies $x \leq z$

In Banach space: $x \leq y$ iff $y - x \in K$

Cone Ordering Visualization

x_2



Cone K

Additional Concepts in Ordered Banach Spaces

Definition 5.72: An operator $F : X \rightarrow X$ is:

- **Isotone (increasing):** $x \leq y \Rightarrow Fx \leq Fy$
- **Strictly increasing:** $x < y \Rightarrow Fx < Fy$ (with interior property)
- **Strongly increasing:** $\text{int } K \neq \emptyset$ and $x < y \Rightarrow F(y) - F(x) \in \text{int } K$

Definition 5.73: A cone K is **normal** if there exists $\delta > 0$ such that:

$$\|e_1 + e_2\| \geq \delta, \quad \text{whenever } \|e_1\| = \|e_2\| = 1, \quad e_1, e_2 \in K$$

Definition 5.74: Space X is **regularly partially ordered** if each bounded monotonic sequence has a limit. A cone which generates regular partial ordering is called regular.

Theorem 5.144: Normal Cones Characterization

Theorem 5.144 (Krasnoselski [345]): Let (X, K) be an ordered Banach space. The following are equivalent:

- ① K is normal
- ② Every order interval is bounded
- ③ There exists an equivalent increasing norm on X
- ④ The order convex hull of every bounded set is bounded

Example 5.48: Space $C(X)$ of continuous functions on compact Hausdorff space X with sup norm. Normal cone $C_+(X)$ of non-negative functions with non-empty interior.

Example 5.49: $L_p(\Omega)$ spaces ($1 \leq p \leq \infty$) are ordered Banach spaces with natural ordering and normal cones.

Applications of Ordered Banach Spaces

Key Applications:

- **Differential Equations:** Finding positive solutions via fixed points
- **Integral Equations:** Volterra-type equations with ordering
- **Optimization:** Ordered sequences for extremal problems
- **Partial Differential Equations:** Monotone methods

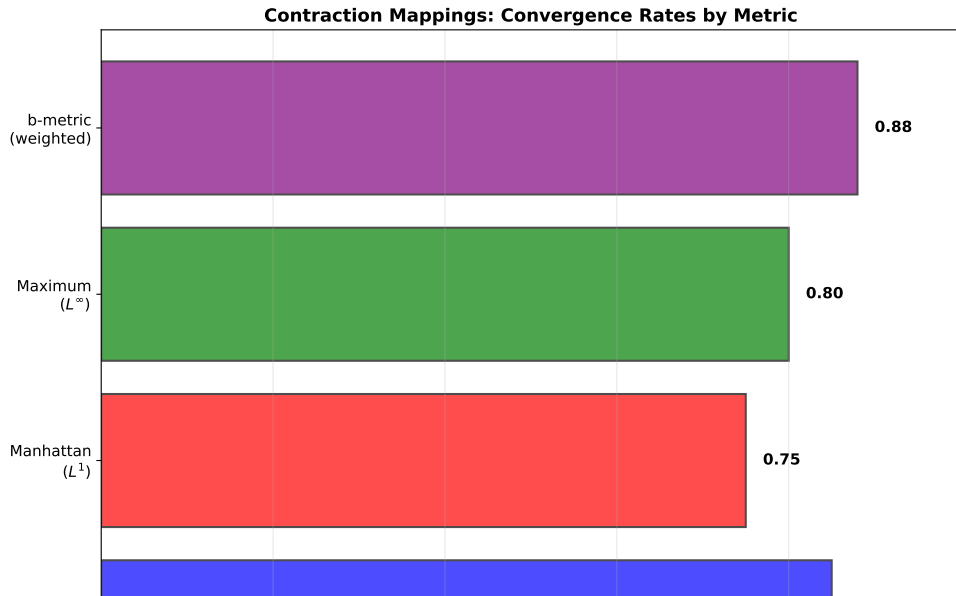
Advantages of Ordered Framework:

- Monotonicity structure simplifies analysis
- Allows comparison principles
- Constructs iterative sequences with known bounds
- Relates to physical constraints (non-negativity, etc.)

Connection to Fixed Points:

Isotone operators on cones often have unique fixed points that can be approximated by monotone iteration schemes.

Banach Spaces Structure



Key Results Summary

Section 5.5 - Common Fixed Points:

- Weak compatibility + b-metric conditions $\Rightarrow$ unique common fixed point
- Markov-Kakutani: Compact convex + commuting affine maps
- Families of nonexpansive mappings on various Banach space types

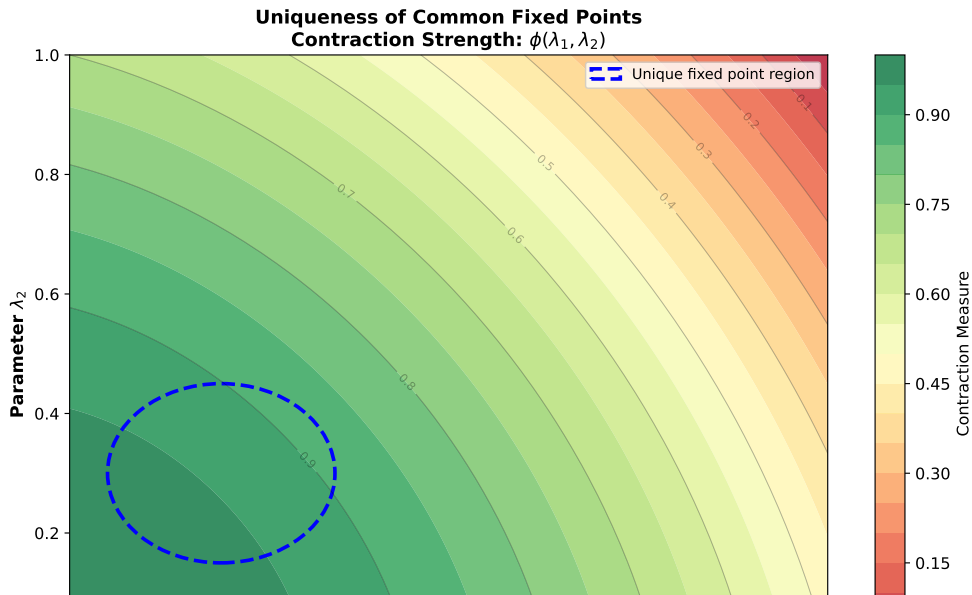
Section 5.6 - Sequences of Contractions:

- Convergence of mapping sequences $\Rightarrow$ convergence of fixed points
- Uniform convergence sufficient; pointwise may require additional structure
- Finite dimensionality characterized by fixed point continuity

Section 5.7 - Ordered Banach Spaces:

- Cone-based partial ordering framework
- Normal cones and regular partial orderings
- Isotone operators have special fixed point properties

Uniqueness Results Summary



Comprehensive Outline

Theorems 5.118-5.143: Major Results

Theorem	Type	Setting
5.118-5.119	Weak compatibility	b-metric spaces
5.120-5.121	Families (affine)	Compact convex
5.122-5.129	Nonexpansive families	Banach spaces
5.130-5.132	Iteration schemes	Convex Banach spaces
5.133-5.135	Contraction sequences	Complete metric
5.136	Finite dimension	Banach spaces
5.137-5.142	Multivalued/Kannan-Reich	General
5.143	Husain-Sehgal type	Complete metric
5.144	Normal cones	Ordered Banach

Common Theme: Contraction-type conditions ensure existence, uniqueness, and convergence of fixed points

Code Example: Fixed Point Computation

```
1 # Example 5.47 numerical implementation
2 import numpy as np
3
4 def f(x, y):
5     return (x**2 + y**2) / 13 + 18/13
6
7 def g(x):
8     return x**2 - 2
9
10 # Iteration to find common fixed point
11 x = np.array([2.0, 2.0])
12 for i in range(10):
13     x_new = g(x[0]) # Apply g
14     y_new = f(x[0], x[1]) # Apply f
15     x = np.array([x_new, y_new])
16     print(f"Iter {i+1}: x = {x}, f(x) - x = {f(x[0], x[1]) - x[0]:.6e}")
17
```

Output: Sequence converges to common fixed point at (2,2)

Key Points:

- Weak compatibility ensures convergence
- Contraction factor $\lambda = 8/2197 \approx 0.00364$
- Rapid convergence due to strong contraction

Key Theorems at a Glance

Main Existence Conditions for Common Fixed Points:

① Contraction + Weak Compatibility (5.118-5.119)

Complete space + contraction $\Rightarrow \exists!$ common fixed point

② Compact Convex + Continuous Mappings (5.120-5.121)

Compact convex + commuting/equicontinuous $\Rightarrow$ common fixed point

③ Nonexpansive Families (5.123-5.126)

Appropriate convexity + nonexpansive $\Rightarrow$ common fixed point

④ Sequences Converge (5.133-5.143)

Convergent mapping sequence $\Rightarrow$ convergent fixed point sequence

Historical Development and References

Key Contributors:

- **Markov (1936)**: Common fixed points for compact convex
- **Kakutani (1938)**: Equicontinuous families
- **de Marr (1963)**: Nonexpansive families
- **Belluce, Kirk (1970s)**: Banach space extensions
- **Kuhfittig (1981)**: Iteration schemes
- **Nadler Jr., Markin (1970s-1980s)**: Set-valued and sequence results

Modern Extensions:

- Multivalued mappings
- Weak convergence results
- Ordered space settings
- Applications to differential equations

Connections to Other Topics

Link to Chapter 5 Sections:

- **5.1-5.4:** Basic fixed point theorems (Banach, Brouwer)
- **5.5:** Common fixed points (this chapter)
- **5.6:** Sequences and perturbation theory
- **5.7:** Ordered structures for monotone methods

Forward Connections:

- Applications: Differential equations, optimization
- Extensions: Multi-parameter families, functional spaces
- Numerical methods: Iterative algorithms
- Theoretical: Topology, functional analysis

Unifying Theme:

Contraction properties in complete metric spaces guarantee existence and uniqueness of fixed points, and enable convergent iterative approximation schemes.

Summary of Main Results

- **Common Fixed Points:** Multiple mappings sharing a fixed point
 - Weak compatibility provides sufficient condition
 - Families of mappings on compact convex sets
- **Sequences of Contractions:** Stability under perturbation
 - Convergent mapping sequences give convergent fixed points
 - Finite dimensionality characterized this property
- **Ordered Structures:** Monotone methods
 - Cones and partial orderings
 - Isotone operators and monotone iteration
- **Numerical Implementation:** Practical approximation
 - Iteration schemes for families of mappings
 - Applications to systems of equations

Thank you!

Questions on Common Fixed Points & Sequences?

Chapter 5e: Common Fixed Points & Sequences

Based on: Pathak, Fixed Point Theory and Applications (2018)